

# Generation Interconnection Combined Feasibility / System Impact Study Report

## For Queue Project Q21-1889

Asheville 138 kV  
December 25, 2022

### **1. Introduction**

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1889. The Interconnection Customer proposes a 1053.5 MW natural gas combined cycle facility interconnecting at the Asheville 138 kV substation in the Virginia area. The request was received on September 26, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 615 higher-queued active requests and their assigned network upgrades.

### **2. Summary of Results**

The project was found to contribute (DFAX of 5% or greater) to 17 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$54,358,000.

### **3. Network Impacts**

#### **3.1 Generator Deliverability and Single Contingency Analysis**

1. Facility: Asheville-Ladysmith Hydro 138 kV Cir. 2  
Contingency: Asheville-Ladysmith Hydro 138 kV  
Loading: 178.4% (Rating: 336 MVA)  
DFAX: 60.5%  
Pre-project loading: 11.2%
2. Facility: Asheville-Ladysmith Hydro 138 kV #1  
Contingency: Ladysmith Hydro Substation-Asheville 138 kV Cir. 2  
Loading: 159.9% (Rating: 375 MVA)  
DFAX: 60.5%  
Pre-project loading: 10.0%
3. Facility: Poages Mill-Floyds Knobs 138 kV L  
Contingency: Asheville-Safeway Stores Substation 138 kV  
Loading: 158.9% (Rating: 223 MVA)  
DFAX: 20.8%  
Pre-project loading: 60.8%
4. Facility: Ladysmith Hydro-Leesville 138 kV L  
Contingency: Floyds Knobs-Poages Mill 138 kV Line  
Loading: 154.8% (Rating: 216 MVA)  
DFAX: 25.2%  
Pre-project loading: 32.1%
5. Facility: Hinton Ap Substation-Oxnard 138 kV L  
Contingency: Central Juncos 3202-Colton 138 kV  
Loading: 116.1% (Rating: 328 MVA)  
DFAX: 6.1%  
Pre-project loading: 96.4%

6. Facility: De Kalb-Oxnard 138 kV  
Contingency: Colton-Central Juncos 3202 138 kV Line  
Loading: 115.4% (Rating: 337 MVA)  
DFAX: 6.1%  
Pre-project loading: 96.1%
7. Facility: McClain Energy Facility-De Kalb 138 kV  
Contingency: Colton-Central Juncos 3202 138 kV  
Loading: 114.8% (Rating: 344 MVA)  
DFAX: 6.1%  
Pre-project loading: 95.9%
8. Facility: Banning-Franklin-Elk Grove 138 kV  
Contingency: Franklin-Elk Grove-S Texa 138 kV  
Loading: 113.7% (Rating: 354 MVA)  
DFAX: 6.1%  
Pre-project loading: 95.4%
9. Facility: Desert Outpost-Banning 138 kV Line  
Contingency: S Texa-Franklin-Elk Grove 138 kV  
Loading: 112.6% (Rating: 369 MVA)  
DFAX: 6.1%  
Pre-project loading: 95.1%
10. Facility: Hancock-Flyds Knobs 138 kV line  
Contingency: Asheville Substation-Safeway Stores 138 kV  
Loading: 112.1% (Rating: 213 MVA)  
DFAX: 18.3%  
Pre-project loading: 21.3%
11. Facility: Littlefork-Lake Lure 138 kV  
Contingency: Fluvanna P.S-Louisa Generation Facility 345 kV line  
Loading: 110.1% (Rating: 918 MVA)  
DFAX: 6.4%  
Pre-project loading: 102.8%
12. Facility: Franklin-Elk Grove-New London 138 kV Line  
Contingency: Bear-Clover 345 kV L  
Loading: 107.4% (Rating: 637 MVA)  
DFAX: 9.0%  
Pre-project loading: 92.4%
13. Facility: South Texa-Franklin-Elk Grove 138 kV line  
Contingency: Desert Outpost-S Texa 138 kV  
Loading: 100.8% (Rating: 611 MVA)  
DFAX: 6.6%  
Pre-project loading: 89.4%
14. Facility: Fluvanna P.S-Louisa Generation Facility 345 kV line  
Contingency: Leavitt-Birchwood 345 kV line  
Loading: 100.6% (Rating: 3,520 MVA)  
DFAX: 17.9%  
Pre-project loading: 95.3%
15. Facility: Louisa Generation Facility-N Anna 345 kV  
Contingency: Carrabelle-Louisa Generation Facility 345 kV  
Loading: 100.3% (Rating: 3,019 MVA)  
DFAX: 12.3%  
Pre-project loading: 96.0%
16. Facility: Bear-Clover Substation 345 kV  
Contingency: Bear-Mecklenburg Pwer Station 345 kV

Loading: 100.1% (Rating: 4,332 MVA)

DFAX: 30.4%

Pre-project loading: 92.7%

17. Facility: Yaphank Substation-Brush Tavern 138 kV Line

Contingency: Glen Oaks-S Texa Substation 138 kV line

Loading: 100.0% (Rating: 507 MVA)

DFAX: 5.4%

Pre-project loading: 88.7%

### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

## 4. Required Network Upgrades and Cost Estimates

| Facility                                            | Required Upgrade                            | Cost Estimate | Allocation to Project |
|-----------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| Asheville-Ladysmith Hydro 138 kV Circuit 2          | Rebuild line with higher-capacity conductor | \$6,774,000   | 80% (\$5,415,000)     |
| Asheville-Ladysmithhydro 138 kV #1                  | Rebuild line with higher-capacity conductor | \$7,780,000   | 80% (\$6,220,000)     |
| Poages Mill-Floyds Knobs 138 kV L                   | Rebuild line with higher-capacity conductor | \$3,759,000   | 72% (\$2,717,000)     |
| Ladysmith Hydro-Leesville 138 kV L                  | Rebuild line with higher-capacity conductor | \$25,719,000  | 85% (\$21,938,000)    |
| Hinton Ap Substation-Oxnard 138 kV L                | Rebuild line with higher-capacity conductor | \$4,423,000   | 12% (\$552,000)       |
| De Kalb-Oxnard 138 kV                               | Rebuild line with higher-capacity conductor | \$8,359,000   | 12% (\$1,044,000)     |
| Mcclain Energy Facility-De Kalb 138 kV line         | Rebuild line with higher-capacity conductor | \$5,203,000   | 12% (\$650,000)       |
| Banning-Franklin-Elk Grove 138 kV                   | Rebuild line with higher-capacity conductor | \$2,086,000   | 18% (\$378,000)       |
| Desert Outpost-Banning 138 kV Line                  | Rebuild line with higher-capacity conductor | \$11,559,000  | 18% (\$2,096,000)     |
| Floyds Knobs-Hanck 138 kV L                         | Rebuild line with higher-capacity conductor | \$4,685,000   | 71% (\$3,306,000)     |
| Littleork-Lake Lure 138 kV                          | Rebuild line with higher-capacity conductor | \$7,314,000   | 12% (\$886,000)       |
| Franklin-Elk Grove-New London 138 kV Line           | Reconductor line                            | \$1,796,000   | 16% (\$297,000)       |
| South Texa-Franklin-Elk Grove 138 kV line           | Reconductor line                            | \$3,751,000   | 16% (\$599,000)       |
| Fluvanna P.S-Louisa Generation Facility 345 kV line | Reconductor line                            | \$12,217,000  | 11% (\$1,377,000)     |
| Louisa Generation Facility-N Anna 345 kV L          | Reconductor line                            | \$34,733,000  | 10% (\$3,542,000)     |

| Facility                                    | Required Upgrade           | Cost Estimate | Allocation to Project |
|---------------------------------------------|----------------------------|---------------|-----------------------|
| Bear-Clover 345 kV L                        | Reconductor line           | \$24,171,000  | 12% (\$3,020,000)     |
| Yaphank Substation-Brush Tavern 138 kV Line | Reconductor line           | \$1,586,000   | 20% (\$321,000)       |
|                                             | Total allocated to project |               | \$54,358,000          |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.